

Michell’s Wave-Resistance Integral and the Free-Wave Amplitude Grid from a B-spline Control Net

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Abstract

This note documents the core numerical calculation performed by the `michell` crate: given a ship hull represented as a tensor-product B-spline half-breadth surface, it computes the thin-ship (Michell) wave resistance and the far-field free-wave elevation pattern. The distinguishing feature of the implementation is that the inner Michell integrals are evaluated *exactly* — in closed form per knot span — because a polynomial B-spline hull makes the integrand a polynomial multiplied by an oscillatory / exponentially decaying kernel on each span. The only numerical error lives in a single smooth one-dimensional outer quadrature. Section references point at the Rust source so the formulae and the code can be read side by side.

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1 Geometry, conventions, and constants

The hull is a slender body symmetric about its centerplane. Its starboard surface is given as a single-valued *half-breadth* (half-beam) function

$$y = f(x, z) \geq 0, \tag{1}$$

where

- x runs along the hull, with the ship advancing toward $+x$ (the bow is the high- x end of the hull file; the wake trails toward $-x$);
- z is measured *downward* from the undisturbed waterline, so the wetted surface occupies $z \in [0, T]$ with T the draft;
- y is the transverse (port/starboard) coordinate.

The surface domain is $x \in [x_0, x_1]$ (length $L = x_1 - x_0$) and $z \in [0, T]$. Only the polynomial (non-rational) B-spline surface of Section 2 is admitted; rational NURBS weights are deliberately excluded, because it is precisely the piecewise-polynomial structure that makes the inner integrals closed-form.

Operating conditions and constants. Table 1 lists the physical inputs. The single derived quantity that governs the whole calculation is the *fundamental wavenumber*

$$\boxed{\nu = \frac{g}{U^2}} \quad (\text{nu in the source}), \quad (2)$$

the wavenumber of the transverse ($\theta = 0$) waves of a steadily advancing ship in deep water. Its reciprocal length scale is the transverse wavelength $\lambda_{\text{tr}} = 2\pi/\nu = 2\pi U^2/g$. The length Froude number $\text{Fr} = U/\sqrt{gL}$ is the usual dimensionless speed.

Symbol	Meaning	Units	Source
U	ship speed	m/s	<code>Conditions::speed</code>
g	gravitational acceleration	m/s ²	9.80665
ρ	fluid density	kg/m ³	<code>Fluid::density</code> (≈ 1025.9 seawater)
ν	fundamental wavenumber g/U^2	1/m	derived, Eq. (2)
L	waterline length	m	$x_1 - x_0$
T	draft	m	z -domain upper bound

Table 1: Physical inputs (`conditions.rs`). Gravity and density carry the ITTC 15°C default values.

2 The B-spline control-net representation

The half-breadth surface is a clamped tensor-product B-spline of degree p in x and q in z :

$$f(x, z) = \sum_{i=0}^{n_x-1} \sum_{j=0}^{n_z-1} P_{ij} N_{i,p}(x) N_{j,q}(z), \quad (3)$$

where $P_{ij} \in \mathbb{R}$ is the *control net* of half-beam values (stored row-major with the z index fastest, `control[i*n_ctrl_z + j]`), and $N_{i,p}$, $N_{j,q}$ are the B-spline basis functions built from the knot vectors $\{\xi_k\}$ (in x) and $\{\zeta_k\}$ (in z). Both knot vectors are *clamped*: the first and last knots each repeat $p + 1$ (resp. $q + 1$) times, so the surface interpolates the corner control points and the parametric coordinate coincides with the physical coordinate. An interior knot of multiplicity up to p is allowed — multiplicity p drops the surface to C^0 and is exactly how a *chine* (hard crease) is represented.

Because the control net is non-negative (a sufficient condition for $f \geq 0$ by the convex-hull property) and $z_0 = 0$ (the domain starts at the waterline), the object is a valid hull. Basis functions and their derivatives are evaluated with the standard Piegl–Tiller recurrences (`bspline.rs`, `ders_basis`).

2.1 From control net to per-span polynomial

The knot vectors partition the domain into a grid of non-empty rectangular *knot spans*. On a single span $[\xi_s, \xi_{s+1}] \times [\zeta_t, \zeta_{t+1}]$ the surface (3) is an ordinary bivariate polynomial of degree (p, q) . The crate materialises that polynomial once, from the mixed partial derivatives evaluated at the span’s lower-left corner $(x_0, z_0) = (\xi_s, \zeta_t)$:

$$D_{ab} = \left. \frac{\partial^{a+b} f}{\partial x^a \partial z^b} \right|_{(x_0, z_0)}, \quad 0 \leq a \leq p, \quad 0 \leq b \leq q, \quad (4)$$

(`BSplineSurface::corner_partials`). Taylor’s theorem then reconstructs the surface exactly on the span,

$$f(x, z) = \sum_{a=0}^p \sum_{b=0}^q \frac{D_{ab}}{a! b!} (x - x_0)^a (z - z_0)^b. \quad (5)$$

Michell’s integral needs the longitudinal slope $\partial f / \partial x$. Differentiating (5) and reindexing gives, on each span, the polynomial

$$f_x(x, z) \equiv \frac{\partial f}{\partial x} = \sum_{a=0}^{p-1} \sum_{b=0}^q c_{ab} (x - x_0)^a (z - z_0)^b, \quad c_{ab} = \frac{D_{a+1, b}}{a! b!}. \quad (6)$$

These coefficients c_{ab} are precomputed for every span pair and stored flattened as `fx_coeff` (`hull.rs`), indexed $[(s_x n_z + s_z) p + a](q + 1) + b$. This array is the sole geometric input to the wave calculation: the entire hull enters Michell’s integral through the per-span slope polynomials (6).

3 Michell’s wave-resistance integral

3.1 The classical form

For a thin ship in steady forward motion in calm, deep water, Michell’s integral gives the wave-making resistance as

$$R_w = \frac{4\rho g^2}{\pi U^2} \int_1^\infty (I(\lambda)^2 + J(\lambda)^2) \frac{\lambda^2}{\sqrt{\lambda^2 - 1}} d\lambda, \quad (7)$$

where the real functions I and J are the cosine and sine parts of the complex *free-wave amplitude function*

$$I(\lambda) + i J(\lambda) = \iint_{\text{hull}} \frac{\partial f}{\partial x} e^{-\nu \lambda^2 z} e^{i \nu \lambda x} dx dz. \quad (8)$$

Here the integration variable $\lambda \geq 1$ is the secant of the wave propagation angle θ ; each λ selects one plane-wave component of the ship’s Kelvin pattern. The kernel of (8) has the two signatures of linear ship-wave theory:

- a horizontal oscillation $e^{i \nu \lambda x}$ with wavenumber $k_x = \nu \lambda$ along the track, and

- a vertical decay $e^{-\nu\lambda^2 z}$ with rate $\kappa = \nu\lambda^2$, the deep-water statement that shorter, more-oblique waves are confined nearer the surface.

(See `michell.rs` module docs; cf. Tuck 1989; Dambrine, Pierre & Rousseaux 2016.)

3.2 Secant substitution

The integrand of (7) has an integrable singularity at $\lambda = 1$. Substituting $\lambda = \sec \theta$ ($d\lambda = \sec \theta \tan \theta d\theta$, $\sqrt{\lambda^2 - 1} = \tan \theta$) sends $\lambda^2/\sqrt{\lambda^2 - 1} d\lambda \rightarrow \sec^3 \theta d\theta$ and removes the singularity, giving the form the code actually integrates:

$$R_w = \frac{4\rho g^2}{\pi U^2} \int_0^{\pi/2} |F(\sec \theta)|^2 \sec^3 \theta d\theta, \quad F(\lambda) \equiv I(\lambda) + i J(\lambda). \quad (9)$$

The prefactor $4\rho g^2/(\pi U^2)$ is `coeff` in `multihull_wave_resistance_with`; the integral is accumulated in `integrate_outer`. This outer integral is smooth, and is the *only* place numerical quadrature error enters.

3.3 Exact evaluation of the inner integral

Substitute the per-span slope polynomial (6) into the definition (8). Because f_x is piecewise polynomial and the kernel separates in x and z , the double integral factorises span-by-span into a sum of products of one-dimensional *moment integrals*. With $k_x = \nu\lambda$ and $\kappa = \nu\lambda^2$, define on a span of lengths h_x, h_z (local coordinates $X = x - x_0 \in [0, h_x]$, $Z = z - z_0 \in [0, h_z]$) the oscillatory and exponential moments

$$M_a(k_x, h_x) = \int_0^{h_x} X^a e^{ik_x X} dX, \quad a = 0, \dots, p-1, \quad (10)$$

$$N_b(\kappa, h_z) = \int_0^{h_z} Z^b e^{-\kappa Z} dZ, \quad b = 0, \dots, q. \quad (11)$$

Then the contribution of a single hull to the amplitude function is the exact finite sum

$$F(\lambda) = \sum_{s \in \text{spans}_x} e^{ik_x(\xi_s - x_c)} \sum_{a=0}^{p-1} M_a(k_x, h_x^{(s)}) \underbrace{\sum_{t \in \text{spans}_z} \sum_{b=0}^q c_{ab}^{(s,t)} e^{-\kappa \zeta_t} N_b(\kappa, h_z^{(t)})}_{g_a^{(s)}}. \quad (12)$$

This is exactly the double loop in `InnerIntegral::eval`: the inner bracket $g_a^{(s)}$ contracts the slope coefficients c_{ab} against the decay-shifted z -moments $e^{-\kappa \zeta_t} N_b$ (precomputed once per λ across all z -spans, array `zm`); the outer sum weights them by the x -moments M_a and the per-span phase factor. The reference phase $x_c = \frac{1}{2}(x_0 + x_1)$ is the hull midpoint; shifting it multiplies F by a unit-modulus factor and leaves $|F|^2$ — hence R_w — invariant.

The upshot: for a polynomial hull, I and J carry *no* discretisation error. Every span integral is a closed form (Section 3.4). The number of inner evaluations equals $(\#x\text{-spans}) \times (\#z\text{-spans}) \times p \times (q+1)$ complex operations per λ .

3.4 Closed-form span moments

The moments (10)–(11) are computed in `moments.rs`. Each has a stable large-argument recurrence obtained by integration by parts, and a small-argument power series used to avoid catastrophic cancellation (switch-over at $|k_x h_x|, \kappa h_z = 4$).

Oscillatory moments. Integration by parts of (10) gives the upward recurrence

$$M_0 = \frac{e^{ik_x h_x} - 1}{ik_x}, \quad M_a = \frac{h_x^a e^{ik_x h_x} - a M_{a-1}}{ik_x}, \quad (13)$$

and for small $|k_x h_x|$ the series

$$M_a = h_x^{a+1} \sum_{m \geq 0} \frac{(ik_x h_x)^m}{m! (a + m + 1)}. \quad (14)$$

Exponential moments. Likewise for (11) with $\kappa \geq 0$,

$$N_0 = \frac{1 - e^{-\kappa h_z}}{\kappa}, \quad N_b = \frac{b N_{b-1} - h_z^b e^{-\kappa h_z}}{\kappa}, \quad (15)$$

and for small κh_z ,

$$N_b = h_z^{b+1} \sum_{m \geq 0} \frac{(-\kappa h_z)^m}{m! (b + m + 1)}. \quad (16)$$

As $\kappa \rightarrow \infty$ (deeply oblique waves) $N_b \rightarrow b!/\kappa^{b+1} \rightarrow 0$, so deep parts of the hull decouple gracefully; the code short-circuits a span once $e^{-\kappa \zeta_t}$ underflows.

3.5 Outer quadrature

The smooth θ -integral (9) is evaluated by marching adaptive Gauss–Legendre panels (16 nodes each). The panel width is set to a fixed fraction of one local oscillation period of the integrand,

$$\Delta\theta \lesssim \frac{2\pi}{r(\theta)}, \quad r(\theta) = 2\nu \sec\theta \tan\theta (x_{1/2} + T \sec\theta) + \dots, \quad (17)$$

where $r(\theta)$ is the analytic phase rate of $|F|^2 \sec^3\theta$ (dominated near the track by the longitudinal bandwidth $x_{1/2}$ and the decay depth T). The pass is repeated with the panels halved until the change between successive passes drops below the requested relative tolerance (`rel_tol`, default 10^{-5}). The tail in $\lambda = \sec\theta$ is truncated once a full multi-period window of the oscillatory integrand contributes negligibly, giving the reported `max_lambda`.

3.6 Fleets and interference

For a multihull / fleet, the far-field free-wave amplitudes superpose linearly. A member hull j at longitudinal offset Δx_j and transverse position y_j contributes $F_j(\lambda) \exp[i\nu(\lambda \Delta x_j \pm \lambda \sqrt{\lambda^2 - 1} y_j)]$ to the wave system propagating at $\pm\theta$ off the track. Writing $k_x = \nu\lambda$ and $k_y = \nu\lambda\sqrt{\lambda^2 - 1}$, the resistance integrand replaces $|F|^2$ in (9) by the mean of the two signed systems,

$$|A(\lambda)|^2 = \frac{1}{2} \left(\left| \sum_j F_j e^{i(k_x \Delta x_j + k_y y_j)} \right|^2 + \left| \sum_j F_j e^{i(k_x \Delta x_j - k_y y_j)} \right|^2 \right), \quad (18)$$

which is `amp_sq` in the source. The two halves differ only for fleets that are not mirror-symmetric about their mean centerplane (e.g. staggered pairs). For two identical hulls a transverse separation s apart this reduces to the classical catamaran interference factor $4 \cos^2(\frac{1}{2}\nu s \lambda \sqrt{\lambda^2 - 1})$.

4 The free-wave amplitude and elevation grid

The same amplitude functions F_j that produce the resistance also produce the visible wave pattern (the "wake"). This is handled in `spectrum.rs`.

4.1 Free-wave spectrum

Far behind the hull the disturbance is a superposition of free plane waves, one per propagation angle $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ measured from the track. Deep-water dispersion pins each component's wavenumber and wavevector at

$$k(\theta) = \nu \sec^2 \theta, \quad \mathbf{k}(\theta) = \nu \sec \theta (1, \tan \theta), \quad (19)$$

and the complex amplitude *density* is

$$A(\theta) = -\frac{2\nu}{\pi} \sec^3 \theta \sum_j \overline{F_j(\sec \theta)} \exp[-i \nu \sec \theta (x_j + y_j \tan \theta)], \quad (20)$$

with $\overline{(\cdot)}$ the complex conjugate and (x_j, y_j) hull j 's midpoint in fleet coordinates (`amp_pair`). The conjugate and the $-\frac{2\nu}{\pi} \sec^3 \theta$ prefactor together fix both magnitude and phase: they follow Tuck, Scullen & Lazauskas (2001, 2002) mapped to this crate's conventions, and reproduces the classical Havelock point-source phase (a bow source's transverse system trails a crest first). Its magnitude is pinned by the free-wave resistance identity, which reproduces (9) term for term:

$$R_w = \frac{1}{2} \pi \rho U^2 \int_{-\pi/2}^{\pi/2} |A(\theta)|^2 \cos^3 \theta d\theta. \quad (21)$$

The integrand of (21), namely $\frac{1}{2} \pi \rho U^2 |A(\theta)|^2 \cos^3 \theta$, is the angular resistance density $dR_w/d\theta$ (`resistance_density`).

4.2 Elevation grid

The surface elevation is the real part of the inverse angular transform

$$\zeta(x, y) = \text{Re} \int_{-\pi/2}^{\pi/2} A(\theta) \exp[i \nu \sec \theta (x + y \tan \theta)] d\theta. \quad (22)$$

This is sampled on an inclusive rectangular grid of $n_x \times n_y$ points spanning $[x_0, x_1] \times [y_0, y_1]$, stored row-major over y (`WaveGrid.zeta[iy*nx + ix]`); a longitudinal or transverse cut is the $n_y = 1$ or $n_x = 1$ special case. The evaluation exploits three structures:

1. **$\pm\theta$ pairing.** F_j depends only on $\sec \theta$, so the two signed systems $A(+\theta)$ and $A(-\theta)$ share their inner-integral evaluations; only the transverse placement phase $e^{\pm i k_y y_j}$ flips. For each $|\theta|$ the row contribution is $A(+\theta)e^{i k_y y} + A(-\theta)e^{-i k_y y}$.
2. **Phasor recurrence in x .** Along a grid row the longitudinal phase $e^{i k_x x}$ (with $k_x = \nu \sec \theta$) advances by the constant factor $e^{i k_x h_x}$ per column, so each column phasor is obtained from the previous by one complex multiply.
3. **Same panel marcher as the resistance.** The θ -integral (22) is integrated with oscillation-rate-sized Gauss–Legendre panels and truncated once the amplitude has decayed (Section 3.5).

Resolution and spectral limits. Two cutoffs keep the grid physical and alias-free:

- *Resolution cutoff.* A grid of spacing (h_x, h_y) can only render components with $\nu \sec \theta h_x < \pi$ and $\nu \sec \theta \tan \theta h_y < \pi$ (at least ~ 2 points per wavelength). The largest resolvable θ is found by bisection, and the last 30% (in $\sec^2 \theta$) before it is cosine-tapered to zero to suppress aliasing. If even the transverse wavelength $2\pi/\nu$ is unresolved, the grid is rejected. Grids limited by this cutoff report `resolution_limited = true`.
- *Spectral cap.* Components beyond $\sec \theta = 15$ (wavelength below $\lambda_{tr}/225$) are always dropped: for wall-sided hulls $|A|$ decays only algebraically toward the track, so the diverging system carries ever-shorter ripple that viscosity would destroy in reality (cf. the eddy-viscosity factor of Tuck, Scullen & Lazauskas 2002).

Validity. Equation (22) is the far-field *free-wave* part of the linear solution. It omits the local (non-radiating) near-field disturbance, so it is the complete linear answer only aft of the stern; on, abreast of, or ahead of the hull it is not physical. Being linear, it also cannot break: steep bow systems are indicative only.

5 Data flow summary

Resistance		
control net P_{ij} , knots	→	B-spline surface f (Eq. 3)
per span: corner partials D_{ab}	→	slope coeffs c_{ab} (Eq. 6, <code>fx_coeff</code>)
c_{ab} , moments M_a, N_b	→	amplitude $F(\lambda) = I + iJ$ (Eq. 12, exact)
$ F ^2$ (or $ A ^2$ for fleets)	→	outer θ -quadrature (Eq. 9)
	→	wave resistance R_w
Wave pattern		
$F_j(\sec \theta)$, placements	→	amplitude density $A(\theta)$ (Eq. 20)
$A(\theta)$, grid (x, y)	→	inverse transform $\zeta(x, y)$ (Eq. 22)
	→	elevation grid <code>WaveGrid</code>

The two branches share the amplitude functions F_j : the resistance integrates $|F|^2 \sec^3 \theta$ over angle, while the wave pattern inverse-transforms $A \propto \bar{F} \sec^3 \theta$ back to physical space. Both are exact in the geometry; the only approximation is the one-dimensional angular quadrature.